Master's Thesis

Reinforcement Learning-Based Feeder Link Switchover for Gateway Load Balancing in NTN

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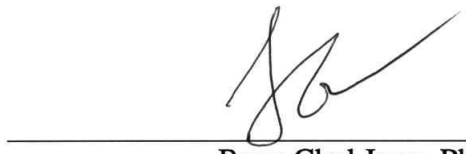
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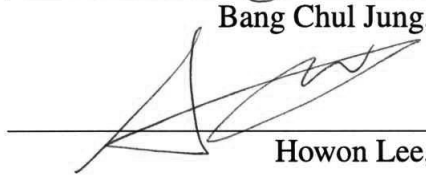
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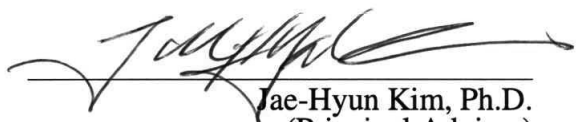
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감사의 글

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Abstract

This thesis proposes a multi-agent deep reinforcement learning (MARL) framework for feeder link switchover (FLSO) to address gateway (GW) load balancing in low Earth orbit (LEO) non-terrestrial networks (NTN). Conventional heuristic FLSO strategies, which rely on simple distance or signal strength, often lead to severe load imbalance across ground GWs and inefficient resource utilization. To address this limitation, this thesis formulates the FLSO decision problem as a decentralized multi-agent Markov decision process (MMDP) where ground GWs act as intelligent agents to jointly optimize link stability, signal quality, and traffic load distribution. This thesis introduces a novel algorithm based on a multi-agent dueling double deep Q-Network architecture, enhanced with prioritized experience replay (PER) and Polyak averaging. The agents learn to make cooperative switchover decisions using locally observable metrics, including satellite ephemeris, L3-filtered reference signal received power (RSRP), and real-time traffic load. The reward function is designed to maximize the network-wide utility by balancing conflicting objectives. Using extensive simulations based on a realistic

Starlink Gen2 constellation and live GW locations, this thesis demonstrates the superiority of the proposed approach. Compared to 3rd generation partnership project (3GPP) heuristics and standard RL baselines, the proposed method achieves the most robust performance trade-off. It effectively suppresses FLSO comparable to conservative heuristics while achieving high link quality and load fairness superior to aggressive signal-based methods. Crucially, it unlocks the network's latent capacity, achieving the highest traffic throughput and connection utilization without causing congestion, thereby validating its potential for efficient autonomous mobility management in next-generation NTN.

Contents

List of Figures **v**

List of Tables **vii**

1 Introduction **1**

1.1 Background and Motivation 1

1.2 Contributions 2

1.3 Overview 3

2 Related Works **4**

2.1 LEO Satellite FLSO Schemes 4

2.2 Load Balancing in LEO Satellite Networks 9

2.3 RL for LEO Mobility Management 10

3 Proposed MARL-based FLSO Algorithm **12**

3.1 System Model 12

3.2 Problem Formulation 16

3.3 Proposed Method 19

3.3.1 MMDP Modeling 19

3.3.2 Proposed Multi-Agent Dueling DDQN Algorithm 21

4 Performance Evaluation **27**

4.1 Simulation Setup 27

4.2 Key Performance Indicators 31

4.3 Results and Discussion 33

5 Conclusion **48**

References **49**

List of Figures

2.1 Packet loss during FLSO process 6

3.1 Network scenario of LEO satellites performing FLSO while across Starlink GWs in the United States. 13
3.2 The architecture of the proposed multi-agent dueling DDQN for FLSO optimization in LEO satellite networks. 23

4.1 Normalized reward with 50 moving average comparison over episodes for the proposed algorithm and ablation baselines (Set 5–8). 35
4.2 Average FLSO per each GW link for the proposed algorithm and baselines across different simulation steps. . . 36
4.3 Average UFLSO per each GW link for the proposed algorithm and baselines across different simulation steps. . . 37
4.4 Average ToS per each GW link for the proposed algorithm and baselines across different simulation steps. 39
4.5 Average RSRP per GW for the proposed algorithm and baselines across different simulation steps. 41

4.6 Average number of connections per GW for the proposed algorithm and baselines across different simulation steps. 42
4.7 Average total traffic load processed per GW for the proposed algorithm and baselines across different simulation steps. 44
4.8 Average Jain's Fairness Index for the proposed algorithm and baselines across different simulation steps. 45

List of Tables

2.1 Entering conditions for measurement events 7

4.1 Simulation parameters 28

List of Abbreviations

3GPP	3rd Generation Partnership Project
AMF	Access and Mobility Management Function
DDQN	Double Deep Q-Network
DRL	Deep Reinforcement Learning
DQN	Deep Q-Network
EIRP	Effective Isotropic Radiated Power
FCC	Federal Communications Commission
FLSO	Feeder Link Switchover
FSPL	Free Space Path Loss
GEO	Geostationary Earth Orbit
GPU	Graphics Processing Unit
GW	Gateway
gNB	Next Generation NodeB
HO	Handover
IIR	Infinite Impulse Response
ISL	Inter-Satellite Link
ITU	International Telecommunication Union

KPI	Key Performance Indicator
L3	Layer 3
LEO	Low Earth Orbit
LoS	Line-of-Sight
MARL	Multi-Agent Reinforcement Learning
mMTC	massive Machine Type Communications
MMDP	Multi-Agent Markov Decision Process
MTS	Minimum Time-of-Stay
NG	Next Generation interface
NLoS	Non-Line-of-Sight
NP	Non-deterministic Polynomial time
NTN	Non-Terrestrial Networks
PER	Prioritized Experience Replay
QoS	Quality of Service
RB	Resource Block
RRM	Radio Resource Management
RRC	Radio Resource Control
RSRP	Reference Signal Received Power
SDN	Software-Defined Networking
TD	Temporal-Difference
ToS	Time-of-Stay
UE	User Equipment
UFLSO	Unnecessary Feeder Link Switchover

Chapter 1. Introduction

1.1. Background and Motivation

Low Earth orbit (LEO) satellite networks have gained significant attention in recent years due to their potential to provide global broadband coverage with low latency. Operating at altitudes between 160 km and 2,000 km above the Earth's surface, LEO satellites offer several advantages over geostationary Earth orbit (GEO) satellites, including reduced signal propagation delay and lower power requirements for communication [1].

However, LEO satellites traveling at approximately 7–8 km/s make it difficult to maintain stable feeder and service links. In transparent-type satellites where the next generation NodeB (gNB) is at the gateway (GW), the satellite performs frequent feeder link switchovers (FLSOs) while passing multiple GWs; when the gNB differs across GWs, each FLSO triggers a simultaneous user equipment (UE) handover (HO) [2]. Such large-scale HOs, especially under 6G massive machine type communications (mMTC), cause signaling bursts that lead to HO failures and degraded service continuity [3]. For regenerative-type satellites, an

on-board gNB allows the UE to maintain its connection directly with the satellite even during a FLSO, meaning a UE HO does not occur as a direct result. However, these satellites still rely entirely on the feeder link for all backhaul traffic. Thus, while they avoid the massive UE HO problem associated with transparent types, the FLSO process itself-namely, selecting the optimal GW for backhaul-remains a critical and frequent operational requirement for maintaining network connectivity. Moreover, heuristic FLSO strategies that prioritize distance-based or signal-based algorithms ignore the processing load and traffic distribution across ground GWs. As a consequence, specific GWs become severely congested while others remain underutilized, amplifying latency, wasting radio resources, and reducing the overall efficiency of the non-terrestrial network (NTN) backhaul.

1.2. Contributions

This thesis proposes a reinforcement learning (RL)-based FLSO scheme that optimizes the trade-off between link quality and GW load balancing. This thesis proposes a dynamic system environment to facilitate the RL agent's training process. The agent leverages locally observable metrics such as reference signal received power (RSRP), LEO satellite ephemeris data, and real-time GW load status. By incorporating these metrics, the agent adapts to rapid orbital dynamics and dynamic traffic, thereby enhancing link stability and reducing network congestion. This thesis designes a multi-objective reward function leveraging RSRP, time-of-stay

(ToS), and GW load to optimize long-term connectivity and network efficiency. This objective function is designed to manage the trade-offs between signal quality, link stability, and load balancing, while minimizing switching instability and load imbalance. This thesis proposes a dueling double deep Q-network-based FLSO algorithm to enhance robustness in noisy and dynamic LEO environments. This algorithm dynamically selects ground GWs to effectively balance network load while simultaneously reducing FLSO frequency, enhancing average RSRP, and extending ToS.

1.3. Overview

The remainder of this thesis is structured as follows: Chapter 2. discusses LEO mobility, FLSO, and related research in load balancing, and RL in LEO NTN. Chapter 3. presents the proposes RL framework in detail. Chapter 4. discusses the simulation results and compares the proposes strategy with conventional approaches. Finally, Chapter 5. concludes the thesis.

Chapter 2. Related Works

2.1. LEO Satellite FLSO Schemes

In NTN, LEO satellites are utilized to provide wide-area coverage while maintaining low end-to-end latency. Since LEO satellites move continuously along their orbits, global service is achieved through multi-satellite, multi-orbit constellations. Mobility management for LEO satellite networks is actively being standardized within 3rd generation partnership project (3GPP). Traditionally, 3GPP specifications focused on UE HO procedures between geographically fixed gNBs, where the next generation (NG) interface toward the core network remained essentially static and did not require dedicated mobility handling. However with LEO satellites, the satellite GW association itself becomes time varying, and the NG interface must track the movement of the satellite. As a result, 3GPP has begun to define explicit NG interface mobility and FLSO procedures for NTN, extending beyond conventional UE HO mechanisms to support this new form of dynamic connectivity [4].

In this context, the satellite payload architecture directly determines how this new mobility management translates into UE HO. For trans-

parent satellites, the gNB is located at the ground GW and UEs are served via the service link. When an FLSO changes the serving GW and the associated gNB, this backhaul level mobility is exposed to the access side, and a corresponding UE HO is triggered.

In contrast, a regenerative satellite has an on-board gNB. In this case, the UE remains anchored to the same gNB even if the feeder link is switched between different GWs, so an FLSO does not inherently require a UE HO. Instead, the impact of mobility is confined to the backhaul domain. Two practical FLSO execution modes are considered: hard FLSO, where the existing feeder link is released before the new one is established, and soft FLSO, where the new feeder link is established prior to releasing the old one.

Moreover, the FLSO process itself introduces significant procedural challenges, which are actively being addressed in 3GPP. As identified in [2], the hard FLSO mechanism can lead to uplink packet loss, as illustrated in Fig. 2.1. This occurs because the access and mobility management function (AMF), unaware of the on-board gNB's instantaneous switch, may continue to forward packets to the old GW after the satellite has already switched to the new GW. These vulnerabilities indicate that not only the execution of FLSO, but also the conditions under which it is triggered, must be carefully controlled to avoid unnecessary service disruption.

In NTN, this control is realized through standardized measurement events that determine when a HO should be initiated [5]. Multiple measurement event conditions are defined to trigger UE HO. For signal-based

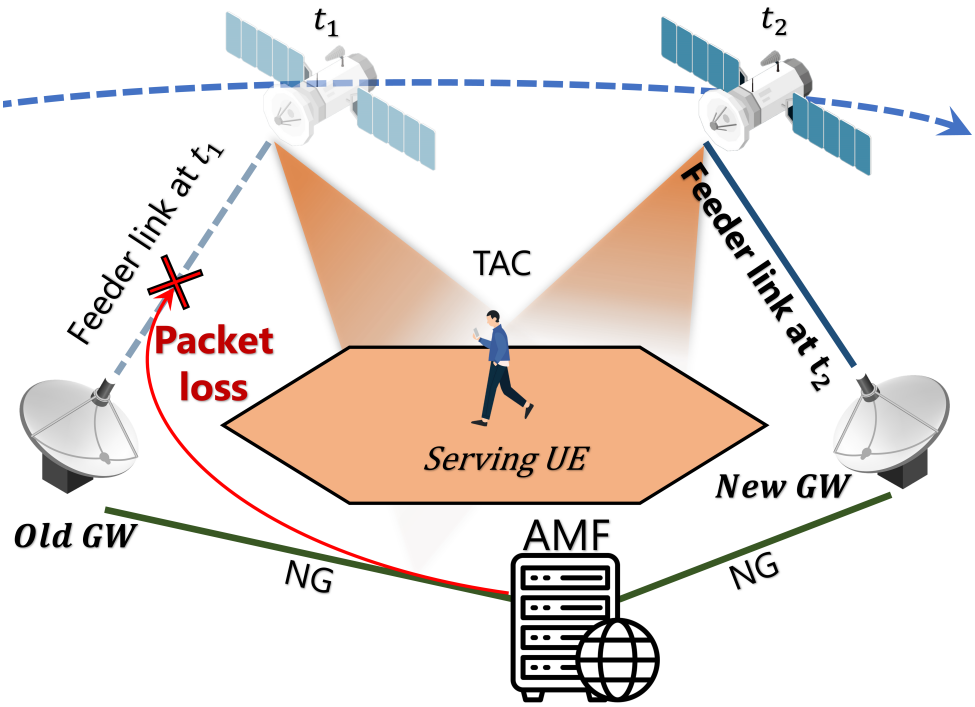


Fig. 2.1. Packet loss during FLSO process

triggers, as summarized in Table 2.1, the serving and target gNBs' RSRPs drive the HO decision; among them, Event A3 is most commonly used. However, for satellites operating at high altitude and moving rapidly, such signal-based events can be unstable. Moreover, on the ground the received signal levels often differ only marginally within a cell radius, making pure signal-based triggering ill-suited to NTN.

Therefore, 3GPP defines location-based measurement events that rely on absolute or moving reference locations, such as cell centers or ephemeris-derived beam centers, together with dual thresholds and hysteresis to mitigate ping-pong behavior [5]. Among these, the dual-threshold Events

Table 2.1. Entering conditions for measurement events

Event	Entering Condition
A1	Serving gNB's signal becomes stronger than the threshold.
A2	Serving gNB's signal becomes weaker than the threshold.
A3	Target gNB's signal becomes stronger than the serving gNB's signal by an Offset margin. (most common HO trigger)
A4	Target gNB's signal becomes stronger than the threshold.
A5	Serving gNB's signal becomes weaker than threshold 1, and Target gNB's signal becomes stronger than threshold 2.

D1 and D2 are most relevant to mobility in LEO NTN scenarios.

Event D1 is specified for static reference locations, such as serving and target cell centers. The triggering condition is evaluated based on the locations from the UE to two fixed reference points, using separate entering and leaving inequalities with a hysteresis margin to suppress rapid toggling.

Event D2 preserves the same decision structure but extends it to moving reference locations determined from satellite ephemeris, for example, time-varying satellite beam centers. This allows the dual-threshold and hysteresis mechanism to operate on dynamically shifting footprints, making D2 directly applicable to LEO NTN, where coverage areas evolve continuously.

Let M_{l1} and M_{l2} denote the locations from the UE to the two reference locations, Thresh_1 and Thresh_2 the corresponding location thresholds, and Hys the hysteresis margin. The generic entering and leaving

conditions can be written as

$$\text{Entering}: \quad (M_{l1} - \text{Hys} > \text{Thresh}_1) \wedge (M_{l2} + \text{Hys} < \text{Thresh}_2), \tag{2.1}$$

$$\text{Leaving}: \quad (M_{l1} + \text{Hys} < \text{Thresh}_1) \vee (M_{l2} - \text{Hys} > \text{Thresh}_2), \tag{2.2}$$

where all locations are measured in meters. Event D1 corresponds to the case in which the reference locations are static, whereas Event D2 applies the same dual-threshold and hysteresis form to ephemeris-based moving references. In both cases, the hysteresis shifts the entering condition outward and the leaving condition inward, ensuring that small fluctuations around the thresholds do not immediately retrigger the event.

In the LEO NTN context, D2-type location-based triggering is particularly attractive because the moving reference locations can track predicted satellite beam centers obtained from broadcast ephemeris information, enabling geometry-aware decisions under rapidly changing topology. However, these standardized location-based events remain essentially link-centric: they stabilize the trigger based on geometric and signal conditions at the UE side, but do not explicitly account for GW processing load or backhaul congestion. As a result, multiple satellites may converge on the same geometrically preferred GW, leading to feeder link bottlenecks. This motivates extending the location-based framework toward a GW-aware FLSO policy that jointly considers location, link stability, and GW load to improve overall NTN backhaul efficiency.

These procedural challenges underscore that FLSO is a complex and costly operation, not a simple link transition. Therefore, an optimal se-

lection policy must not only consider link quality but also minimize the frequency of unnecessary FLSOs.

While 3GPP defines the architectures for executing an FLSO, the optimal selection policy—determining which available GW to switch to—is non-standardized and remains a critical challenge for network performance.

2.2. Load Balancing in LEO Satellite Networks

Load balancing is a critical challenge in LEO satellite networks to ensure overall system efficiency and quality of service (QoS). The time-varying topology of the LEO constellation and the uneven global traffic demand can cause severe congestion on specific satellite nodes or links.

To address this challenge, much of the existing research has been concentrated on load balancing from a routing perspective via inter-satellite links (ISLs) [6–8]. These studies aim to manage the horizontal traffic flow between satellites to avoid congestion on the space-based backbone. These ISL-centric approaches include techniques such as SDN-based adaptive routing [6, 8] and back-pressure routing [7]. For instance, Roth et al. [6] propose a distributed software-defined networking (SDN)-based routing scheme that divides the constellation into clusters and performs load balancing for ISL paths within the cluster. Similarly, Deng et al. [7] propose a Distance-Based Back-Pressure Routing strategy, which calculates ISL link weights based on both the queue backlogs of satellite nodes and the propagation delay, thereby balancing traffic loads among

satellite nodes.

Another domain of research focuses on radio resource management (RRM) at the service link level. These studies aim to balance the load among beams by efficiently allocating limited radio resources, such as beams, spectrum, or time slots, to UEs on the ground.

However, existing load balancing research, such as the ISL-centric routing schemes discussed above [6–8], is predominantly focused on balancing load between satellite ISLs or at the service link RRM level. In contrast, load balancing at the feeder link level, where the satellite constellation as a whole connects to the ground network, remains a relatively unexplored area. Even if congestion at the satellite or ISL level is avoided, the terrestrial network bottleneck persists if all traffic ultimately converges on a single, overloaded GW.

As reviewed in Section 2.1, conventional FLSO strategies tend to rely solely on link-level metrics like RSRP or location. Because these methods do not consider the real-time processing load or traffic status of the GWs, they fail to perform intelligent load balancing. This thesis addresses this gap by integrating GW load awareness into the FLSO decision-making process to optimize the overall efficiency of the NTN backhaul.

2.3. RL for LEO Mobility Management

The agent learns control policies through interaction with an environment: at each decision step the agent observes a state, selects an action, receives a scalar reward, and transitions to the next state [9]. The objec-

tive is to learn a policy that maximizes the discounted long-term return.

Value-based approaches approximate the action–value function $Q(s, a)$. A deep Q-network (DQN) predicts $Q(s, a)$ and selects the action with the highest predicted value. To improve stability and accuracy, common enhancements include double DQN (DDQN), which decouples action selection and evaluation, dueling architecture, which separates value and advantage streams, prioritized replay, which samples transitions with large temporal difference (TD) errors more often, and Polyak averaging for smooth target updates. Function approximators are kept lightweight yet expressive, typically multi-layer perceptrons with normalization and nonlinear activations. Feasibility is enforced by action masking, so that invalid actions are eliminated before scoring. Parameters are updated via backpropagation using Adam or AdamW. Exploration in value-based agents follows an ϵ-greedy schedule, whereas policy-based methods rely on entropy regularization that is gradually annealed. Experience replay breaks temporal correlations and increases sample reuse, and the Huber (smooth-L_1) loss is preferred for TD regression due to its robustness to outliers and non-stationary targets.

Recent studies [10, 11] have applied deep reinforcement learning (DRL) to NTN mobility subproblems, including service link HO and beam resource scheduling, reporting improved throughput and reduced delay compared with heuristic baselines. However, most designs assume centralized observability or ignore the coupling to the feeder link, and thus do not address GW selection or backhaul load balancing explicitly.

Chapter 3. Proposed MARL-based FLSO Algorithm

3.1. System Model

As in Fig. 3.1, this thesis considers a large-scale LEO satellite constellation deployed across P orbital planes, represented by the set $\mathcal{P} = \{p \mid p = 1, \ldots, P\}$. The constellation consists of N LEO satellites in total, distributed across these planes, with N_p satellites in plane p. The overall satellite index set is denoted as $\mathcal{S} = \{s_{p,i} \mid p = 1, \ldots, P;\ i = 1, \ldots, N_p\}$, and its cardinality as $|\mathcal{S}| = \sum_{p=1}^{P} N_p$. In this proposed framework, the ground GWs are modeled as intelligent agents in a multi-agent reinforcement learning (MARL) setting. The set of agents corresponds to the set of GWs $\mathcal{G} = \{1, \ldots, M\}$. Each GW agent m manages its limited number of connection links, denoted as l_{max}, to serve multiple LEO satellites simultaneously. The GW monitors the link conditions of connected satellites and visible candidate satellites to make load balancing and FLSO decisions.

To address the primary challenge of GW load balancing, where the instantaneous load $|\mathcal{L}_m(t)|$ is defined by the set of connected satellites, this

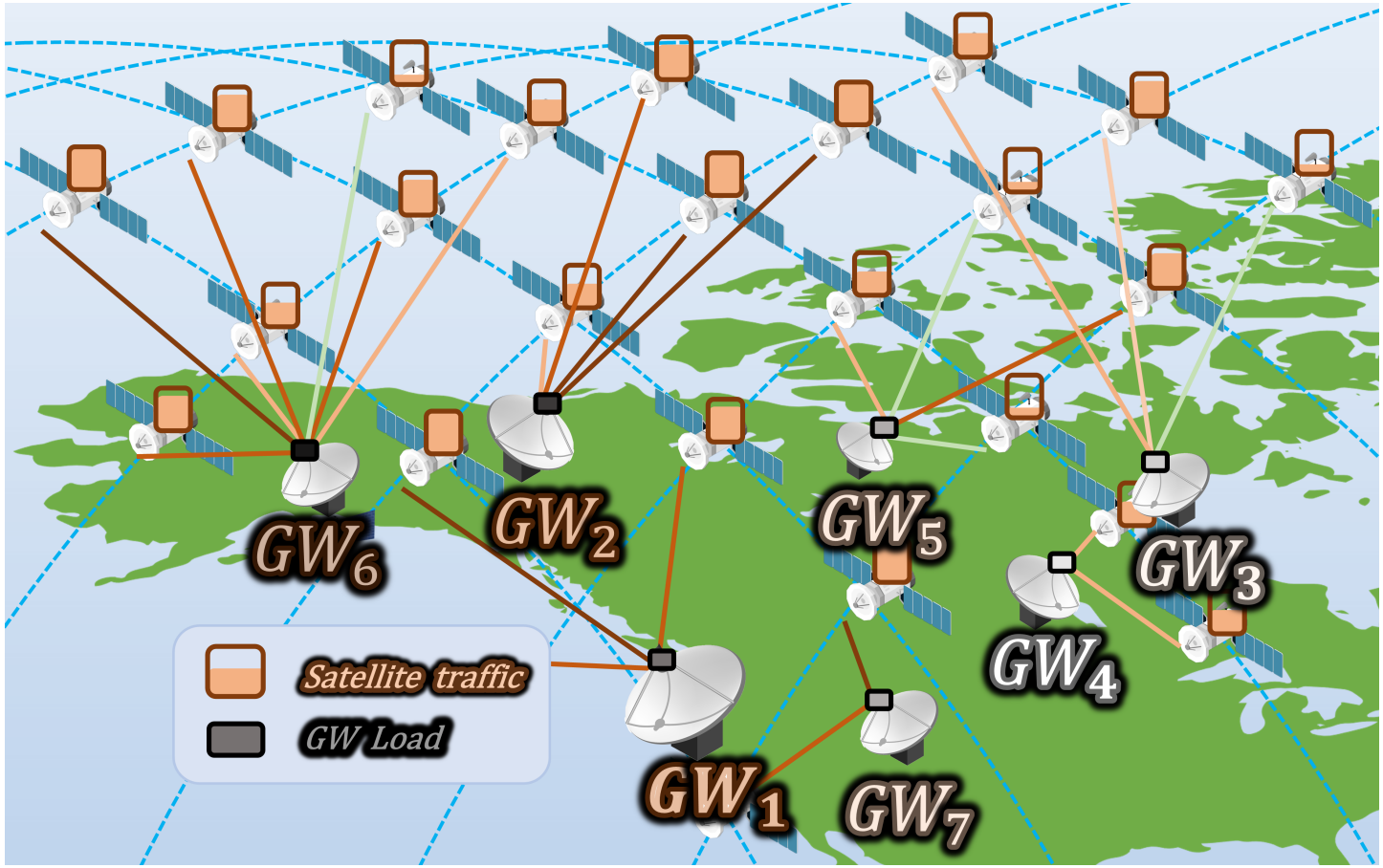


Fig. 3.1. Network scenario of LEO satellites performing FLSO while across Starlink GWs in the United States.

thesis models the dynamic UE demand of each satellite n as a stochastic time-varying traffic load $\psi_n(t)$. Thereby introducing network uncertainty that necessitates GW agents to adaptively manage connections based on fluctuating aggregate traffic rather than solely on satellite visibility.

The GW m receives measurement reports from connected LEO satellites. The satellite n of GW m RSRP $R_{m,n}$ is calculated as follows:

$$R_{m,n} = EIRP_m + G_n - PL_{m,n} - 10\log_{10}(q_{RB}), \tag{3.1}$$

where $EIRP_m$ is the effective isotropic radiated power (EIRP) of the GW m, defined as $EIRP_m = P_m + G_m$, with P_m being the transmit power and G_m the transmit antenna gain of the GW m towards the LEO satellite n. G_n represents the receive antenna gain of the LEO satellite n. The term $10\log_{10}(q_{RB})$ is used to normalize the RSRP measurement, accounting for a single resource block out of the total resource blocks q_{RB} allocated to the link. $PL_{m,n}$ is the large-scale path loss between the GW m and the LEO satellite n, and this expression is adapted from 3GPP documentation [12]. The model is calculated as follows:

$$\begin{aligned} PL_{m,n} &= P_{LOS,m,n}\big(FSPL_{m,n}(d_{m,n}, f_m) + SF_{m,n}\big) \\ &\quad + P_{NLOS,m,n}\big(FSPL_{m,n}(d_{m,n}, f_m) + SF_{m,n} + CL_{m,n}\big) + AT_{m,n}, \end{aligned} \tag{3.2}$$

where $P_{LOS,m,n}$ and $P_{NLOS,m,n}$ denote the probabilities that the LEO satellite n to GW m is in a line-of-sight (LoS) and non-line-of-sight (NLoS) condition, respectively. $FSPL_{m,n}(d_{m,n}, f_m)$ denotes the free space

path loss (FSPL) as a function of the distance $d_{m,n}$ between the LEO satellite n and the GW m and the frequency f_m of the GW m uplink feeder link frequency. $SF_{m,n}$ represents the shadow fading, which accounts for the signal attenuation caused by obstacles such as buildings, trees, or terrain variations in the propagation path. $CL_{m,n}$ denotes the clutter loss, reflecting additional signal degradation due to environmental clutter. $AT_{m,n}$ captures atmospheric attenuation along the feeder link slant path, including gaseous absorption due to dry air and water vapor, and it can be obtained from standard international telecommunication union (ITU)-R models for the considered band [13]. Specifically, the $FSPL_{m,n}$ is calculated as follows:

$$FSPL_{m,n} = 32.45 + 20\log_{10}(f_m) + 20\log_{10}(d_{m,n}). \tag{3.3}$$

At each measurement instant t, the LEO satellite n samples the RSRP from GW m and all neighboring GWs within instantaneous visibility region, yielding the raw RSRP $R_{m,n}(t)$. To mitigate the effects of shadow fading and measurement noise, these raw RSRP $R_{m,n}(t)$ samples are processed at the radio resource control (RRC) layer using a layer 3 (L3) infinite impulse response (IIR) filter [5]. Denoting the filtered RSRP by $\widetilde{R}_{m,n}(t)$, the update equation is calculated as follows:

$$\widetilde{R}_{m,n}(t) = aR_{m,n}(t) + (1-a)\widetilde{R}_{m,n}(t-1), \tag{3.4}$$

where $a \in (0,1)$ is the L3 forgetting factor, defined as $a = \left(\frac{1}{2}\right)^{K/4}$, and K

is the filter coefficient. The initial condition is set as $\widetilde{R}_{m,n}(0) = R_{m,n}(0)$. Finally, the LEO satellite compiles a measurement report containing the L3-filtered RSRP $\widetilde{R}_{m,n}(t)$ of the serving and neighboring GWs and transmits this report back to the serving GW m over the feeder link.

3.2. Problem Formulation

This section addresses the optimization problem for the LEO satellite constellation. The objective is to find an optimal FLSO policy π that dynamically assigns each satellite n to a GW m at each time slot t.

The primary goal is to maximize the network-wide utility, which involves minimizing load imbalance and switching instability while maximizing link quality and connection duration. Unlike simple connection counting, this thesis defines the instantaneous traffic load of GW m at time t, denoted as $|\mathcal{L}_m(t)|$, as the aggregate traffic demand of all connected satellites:

$$|\mathcal{L}_m(t)| = \sum_{n \in \mathcal{S}_m(t)} |\psi_n(t)|, \tag{3.5}$$

where $\mathcal{S}_m(t)$ denotes the set of connected satellites served by GW m at time t and $|\psi_n(t)|$ is the dynamic traffic demand of satellite n.

To align with the MARL framework, this thesis formulates the problem as a maximization of cumulative rewards. The following reward and penalty components are defined:

- **FLSO Penalty:** A penalty component C_{FS} is incurred if the satel-

lite switches its serving GW, penalizing instability.

$$FS_{m,n}(t) = \begin{cases} C_{FS} & \text{if } m_n(t) \neq m_n(t-1), \\ 0 & \text{otherwise,} \end{cases} \tag{3.6}$$

where $m_n(t)$ denotes the index of the serving GW connected to satellite n at time t.

- **UFLSO Penalty:** A penalty component C_{UFS} is incurred for an unnecessary FLSO (UFLSO), which occurs if the satellite switches to another GW within a minimum time-of-stay (MTS), denoted τ_{min}. Let $\tau_{stay}(t)$ be the time elapsed since the last FLSO.

$$UFS_{m,n}(t) = \begin{cases} C_{UFS} & \text{if } m_n(t) \neq m_n(t-1) \text{ and } \tau_{stay}(t) < \tau_{min}, \\ 0 & \text{otherwise.} \end{cases} \tag{3.7}$$

- **ToS Reward:** A reward is gained for maintaining a connection with the same GW, proportional to the duration of the connection.

$$ToS_{m,n}(t) = \tau_{stay}(t). \tag{3.8}$$

- **Fairness Reward:** To ensure load balancing based on the stochastic traffic $\psi_n(t)$, a reward is derived from Jain's Fairness Index [14]

applied to the GW traffic loads $|\mathcal{L}_m(t)|$.

$$J(t) = \frac{(\sum_{m=1}^{M} |\mathcal{L}_m(t)|)^2}{M \cdot \sum_{m=1}^{M} (|\mathcal{L}_m(t)|)^2}. \tag{3.9}$$

- **Link Quality Reward:** A reward is gained based on the L3-filtered RSRP from the connected GW to ensure high-quality connectivity.

The global optimization problem **P1** is to find the policy $\boldsymbol{\pi}$ that maximizes the weighted sum of these components over the time horizon T:

$$\textbf{P1:} \quad \max_{\boldsymbol{\pi}} \sum_{t=0}^{T} \Big[J(t) + \sum_{m=1}^{M} \sum_{n \in \mathcal{S}_m(t)} \Big(\widetilde{R}_{m,n}(t) + ToS_{m,n}(t) - \xi_{m,n}(t) \Big) \Big], \tag{3.10}$$

s.t.

$$\textbf{C1}: \quad \theta_{m,n}(t) \geq \theta_{\min}, \quad \forall m, \forall n \in \mathcal{S}_m(t), \forall t, \tag{3.11}$$

$$\textbf{C2}: \quad \widetilde{R}_{m,n}(t) \geq R_{\min}, \quad \forall m, \forall n \in \mathcal{S}_m(t), \forall t, \tag{3.12}$$

$$\textbf{C3}: \quad |\mathcal{S}_m(t)| \leq l_{\max}, \quad \forall m, \forall t. \tag{3.13}$$

In **P1**, $\xi_{m,n}(t) = FS_{m,n}(t) + UFS_{m,n}(t)$ represents the combined switching penalties. The positive signs for $J(t)$, $\widetilde{R}_{m,n}(t)$, and $ToS_{m,n}(t)$ indicate maximization of fairness, link quality, and stability, respectively, while the negative sign for $\xi_{m,n}(t)$ enforces the minimization of switching overheads. The constraints define the feasibility conditions for the active connections $\mathcal{S}_m(t)$. Constraint **C1** ensures that the elevation angle $\theta_{m,n}(t)$

for every connected link exceeds the minimum threshold $\theta_{\min}$, guaranteeing a reliable LoS. Constraint **C2** requires that the L3-filtered RSRP $\widetilde{R}_{m,n}(t)$ of any active connection maintains a minimum signal quality level $R_{\min}$. Constraint **C3** imposes a hard capacity limit on the ground segment, ensuring that the number of satellites served by a single GW m does not exceed the maximum link capacity $l_{\max}$.

The FLSO optimization problem involves simultaneous decision-making by multiple satellites, characterized by high-dimensional state spaces and complex, time-varying network topology. This combinatorial nature renders the problem non-deterministic polynomial-time (NP) hard.

3.3. Proposed Method

To address the problem **P1**, the optimization problem is formulated as a multi-agent Markov decision process (MMDP) [15]. This framework allows each GW agent to learn an optimal policy for managing satellite connections through sequential interaction with the dynamic LEO environment.

3.3.1 MMDP Modeling

In the proposed framework, each GW m acts as an independent agent. The agent observes its local state $s_{m,t}$, executes an action $a_{m,t}$, and receives a reward $r_{m,t}$. The detailed definitions are as follows:

The state $s_{m,t}$ is designed to provide the agent with comprehensive information about its current utilization and potential handover targets.

It consists of two feature vectors:

$$s_{m,t} = \left\{ \mathbf{H}_{m,t}^{links}, \mathbf{H}_{m,t}^{cands} \right\}, \tag{3.14}$$

where, $\mathbf{H}_{m,t}^{links}$ denotes the status of the agent's l_{max} connection links. For each link l, the feature vector contains a binary active feeder link indicator, the normalized RSRP of the connected satellite, the normalized ToS of the current connection, and the satellite's traffic demand. $\mathbf{H}_{m,t}^{cands}$ encodes the top-K visible candidate satellites sorted by signal strength, where each candidate is represented by its normalized RSRP and traffic demand. This compact, unified representation allows the agent to evaluate and compare its current connections against potential alternatives when making connection and FLSO decisions.

The action space $a_{m,t}$ is discrete and defined as the selection of a candidate satellite to serve. At each step t, the agent selects an index p from the candidate list:

$$a_{m,t} \in \{0, 1, \ldots, p_{cand}\}, \tag{3.15}$$

where p_{cand} is the number of considered candidates. Selecting a candidate index implies attempting to establish a connection with that satellite. If the candidate is already connected, the action is treated as maintaining the connection. If the candidate is new and the GW has available links, a new connection is formed. If the GW is full, the agent may replace an existing connection based on the policy.

The reward function $r_{m,t}$ is engineered to guide the agent towards maximizing network utility while balancing load and minimizing instability. Based on the objective function in **P1**, the immediate reward is calculated as:

$$\begin{aligned} r_{m,t} =& w_r R_m(t) + w_t T_m(t) + w_c C_m(t) + w_F J(t) \\ & - w_s \mathbb{I}(FLSO_m(t)) - w_l (|\mathcal{L}_m(t)|)^2, \end{aligned} \tag{3.16}$$

where, $R_m(t)$ denotes the average normalized RSRP of satellites connected to GW m. $T_m(t)$ is the average normalized ToS promoting connection stability. $C_m(t)$ denotes the average traffic throughput served by GW m. $J(t)$ is Jain's fairness index defined above. $\mathbb{I}(FLSO_m(t))$ is an indicator function equal to 1 when an FLSO occurs at GW m at time t. $|\mathcal{L}_m(t)|$ denotes the total traffic load of GW m the squared term imposes a quadratic penalty on load concentration. The weights $w_r, w_t, w_c, w_F, w_s, w_l$ control the relative importance of these objectives, and the discount factor $\gamma \in [0, 1]$ determines the emphasis placed on future rewards.

3.3.2 Proposed Multi-Agent Dueling DDQN Algorithm

To solve the formulated MMDP, this thesis proposes a multi-agent dueling DDQN algorithm. Unlike traditional single-agent approaches where satellites make independent FLSO decisions, the proposed framework

employs ground GWs as intelligent agents that learn to cooperatively manage traffic loads.

This thesis utilizes a parameter sharing strategy where all homogeneous GW agents share a single Q-network parameterized by θ. This approach significantly reduces the number of trainable parameters, accelerates convergence by aggregating experiences from all agents, and ensures scalability regardless of the number of GWs.

As depicted in Fig.3.2, the architecture incorporates a dueling network structure to separate state-value and advantage estimation, and a DDQN mechanism to mitigate overestimation bias. The shared Q-function $Q(s_{m,t}, a_{m,t}; \theta)$ takes the local state of GW m and its action as input. The loss function is defined as the expectation over experiences sampled from a shared replay buffer:

$$L(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{R}} \left[L_{\text{Huber}}(y_t - Q(s, a; \theta)) \right], \tag{3.17}$$

where the Huber loss is given by:

$$L_{\text{Huber}}(x) = \begin{cases} \frac{1}{2}x^2 & \text{if } |x| \leq 1 \\ |x| - \frac{1}{2} & \text{otherwise} \end{cases}, \tag{3.18}$$

and the target value y_t is calculated using the target network θ^-:

$$y_t = r + \gamma Q(s', \arg\max_{a'} Q(s', a'; \theta); \theta^-). \tag{3.19}$$

where θ^- represents the parameter of the target network. The target

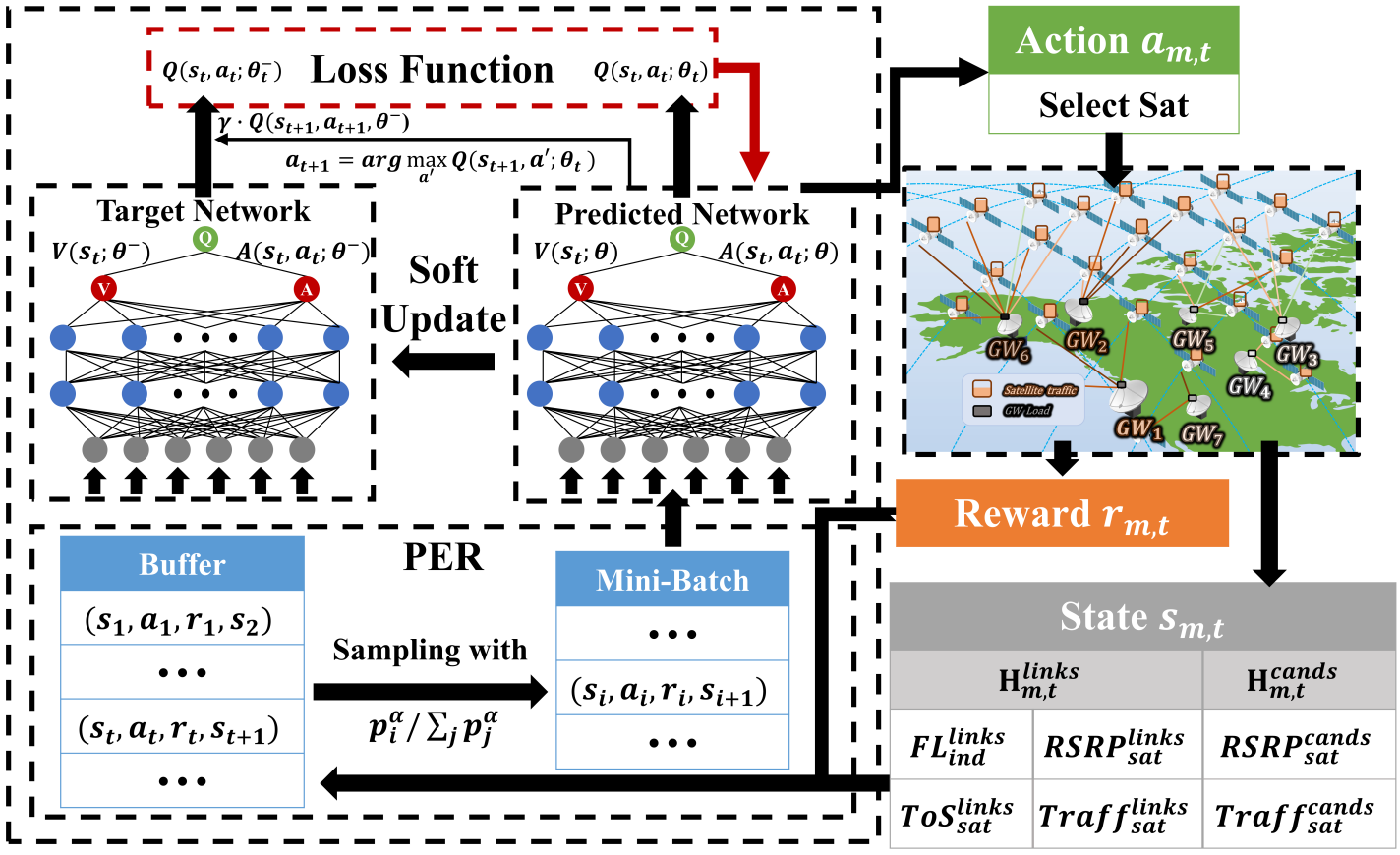


Fig. 3.2. The architecture of the proposed multi-agent dueling DDQN for FLSO optimization in LEO satellite networks.

network parameter are updated as $\theta^- \leftarrow \tau \cdot \theta + (1-\tau) \cdot \theta^-$, ensuring slow and stable updates to mitigate the moving target problem in the dynamic LEO satellite environment [16].

To improve robustness against random noise, such as channel shadow fading, this thesis utilizes a dueling DQN [9]. The dueling DQN architecture enhances the traditional DQN by splitting the Q-function into two separate streams: the state-value function $V(s;\theta)$ and the advantage function $A(s,a;\theta)$. The Q-function is then calculated as follows:

$$Q(s,a;\theta) = V(s;\theta) + \left(A(s,a;\theta) - \frac{1}{|A|} \sum_{a'} A(s,a';\theta) \right), \tag{3.20}$$

where $V(s;\theta)$ represents the value of the state s, $A(s,a;\theta)$ represents the advantage of taking action a in state s, and $|A|$ is the number of possible actions. This decomposition allows the network to better distinguish the value of a state from the advantage of each action, improving robustness against noise and enhancing learning efficiency in complex LEO satellite environments.

To further enhance the learning efficiency, this thesis incorporates prioritized experience replay (PER) as proposed in [17]. PER prioritizes experiences with higher TD errors, where the TD error δ_i for transition i is defined as the difference between the target and current Q-values. The priority is calculated as follows:

$$p_i = |\delta_i| + \epsilon_p, \tag{3.21}$$

where $\delta_i = r_i + \gamma Q(s_{i+1}, a_{i+1}; \theta^-) - Q(s_i, a_i; \theta)$ is the TD error, and ϵ_p ensures non-zero priorities. PER uses $\alpha \in [0, 1]$ to control sampling probability p_i^α and $\beta \in [0, 1]$ to adjust importance sampling weights $(\mathcal{D} \cdot P(i))^{-\beta}$, where $\mathcal{D}$ is replay buffer limit and $P(i)$ is the probability of sampling transition i. This approach accelerates learning by focusing on significant transitions, which is beneficial in LEO satellite networks with dynamic and noisy conditions.

By combining dueling network with DDQN and PER, this approach effectively addresses overestimation bias, accelerates learning, and mitigates the effects of random noise, leading to improved network performance with effective GW load balancing, elevated RSRP, and reduced FLSO frequency. The detailed process of the training is described in Algorithm 3.1.

Algorithm 3.1 Multi-Agent Dueling DDQN-based FLSO

Require: Replay buffer $\mathcal{D}$, batch size $\mathcal{Z}$, episodes $\mathcal{E}$, discount γ, PER parameters $(\alpha, \beta, \epsilon_p)$, exploration ϵ, Polyak τ, learning rate η

Ensure: Optimal decentralized policy π for GW agents

Initialize: Shared online network θ, target network $\theta^- \leftarrow \theta$, Replay buffer $\mathcal{R}$
for *Episode* $= 1$ to $\mathcal{E}$ **do**
  Reset environment, observe initial states $s_{m,0}$ for all $m \in \mathcal{G}$
  for $t = 0$ to T **do**
    Decentralized Action Selection:
    for each agent m **do**
      $$a_{m,t} = \begin{cases} \arg\max_{a'} Q(s_{m,t}, a'; \theta) & \text{w.p. } 1 - \epsilon \\ \text{random action} & \text{w.p. } \epsilon \end{cases}$$
    end for
    Execute joint actions $\mathbf{a}_t$, get rewards $r_{m,t}$ and next states $s_{m,t+1}$
    Store Transitions:
    for each agent m **do**
      Store $(s_{m,t}, a_{m,t}, r_{m,t}, s_{m,t+1})$ into $\mathcal{R}$
    end for
    if $|\mathcal{R}| \geq \mathcal{Z}$ **then**
      Sample minibatch using PER:
      for $i = 1$ to $\mathcal{Z}$ **do**
        Sample $(s_i, a_i, r_i, s_i') \sim \mathcal{R}$ with $P(i) \propto p_i^\alpha$
        $w_i \leftarrow (\mathcal{D} \cdot P(i))^{-\beta}$, normalize w_i
        Compute Target & TD Error:
        $y_i \leftarrow r_i + \gamma\, Q(s_i', \arg\max_{a'} Q(s_i', a'; \theta); \theta^-)$
        $\delta_i \leftarrow y_i - Q(s_i, a_i; \theta)$
        Update priority $p_i \leftarrow |\delta_i| + \epsilon_p$
      end for
      Update Network:
      $L \leftarrow \frac{1}{\mathcal{Z}} \sum_{i=1}^{\mathcal{Z}} w_i\, L_{\text{Huber}}(\delta_i)$
      $\theta \leftarrow \theta - \eta \nabla_\theta L$
      $\theta^- \leftarrow \tau\, \theta + (1 - \tau)\, \theta^-$
    end if
  end for
end for

Chapter 4. Performance Evaluation

4.1. Simulation Setup

In this section, the performance of the proposed algorithm is evaluated alongside conventional baseline schemes, including signal-based, distance-based, and random selection algorithms, through extensive simulations. Furthermore, an ablation study is conducted to validate the effectiveness of specific components integrated into the proposed framework. The simulation environment considers the actual deployment of GWs and satellites. Specifically, it is based on operational SpaceX Starlink GW sites in the United States and models the partially approved Starlink Gen2 satellite constellation by the Federal Communications Commission (FCC) [18, 19]. As the satellites pass over the GWs, the RSRP is periodically measured. The simulation parameters are summarized in Table 4.1. All algorithms share the same physical layer and mobility assumptions to ensure a fair comparison. The simulations are implemented in Python, with 100 independent episodes to capture the statistical behavior of LEO satellite mobility and GW association. While the training phase was conducted with a baseline of 100 steps per episode, the evaluation phase ex-

Table 4.1. Simulation parameters

Environment Parameter [2, 18]	**Value**
Number of GWs M	46
Max connections per GW l_{max}	6
Number of candidate satellites p_{cand}	30
Constellation type	Walker-delta
LEO satellite altitude h	530 [km]
Inclination i	43°
Number of LEO satellites N	3,360
Orbit planes P	28
Traffic load initialization	$\mathcal{U}(0, 1)$ [Gbps]
Traffic model noise	$\mathcal{N}(0, 0.05)$ [Gbps]
Carrier frequency f_m	28 [GHz]
GW Tx power P_m	47 [dBm]
GW Tx antenna gain G_m	49.5 [dBi]
LEO satellite Rx antenna gain G_n	38.5 [dBi]
Ambient temperature T_0	150 [K]
Antenna temperature T_a	290 [K]
Noise figure N_f	3 [dB]
Atmospheric attenuation $AT_{m,n}$	0.02 [dB/km]
Boltzmann constant k	1.379×10^{-23} [J/K]
Number of resource blocks q_{RB}	6
Minimum elevation angle θ_{min}	25°
Minimum GW RSRP R_{min}	-110 [dBm]
FLSO Parameter	**Value**
Measurement period T_m	2 [s]
MTS	20 [s]
L3 filter coefficient K	4
Distance thresholds Δd_{offset}	0, 30 [km]
A3 event ΔR_{offset}	2, 4 [dB]
RL Parameter	**Value**
Learning rate	0.0001
Discount factor γ	0.99
Replay buffer limit $\mathcal{D}$	10^5
Batch size $\mathcal{Z}$	512
Total steps per episode	100
Polyak coefficient τ	0.01
PER exponent α	0.6
PER initial β	0.4

tended the episode duration up to 500 steps to verify the scalability and robustness of the proposed approach. The trained dueling DDQN requires only a few fully connected layers, resulting in sub-millisecond inference time on NVIDIA GeForce RTX 3080 GPU, which is compatible with the 2 second measurement periodicity. Ideally, the performance of the proposed algorithm should be benchmarked against the global optimal solution. However, finding the exact global optimum for the joint FLSO problem across the entire constellation is computationally intractable. The problem involves assigning satellites N to GW M at each time step, subject to connectivity constraints and load balancing objectives. This corresponds to a combinatorial optimization problem with a search space size of approximately $O(M^N)$ per step. Given the large-scale network configuration $N = 3360$ satellites, $M = 46$ GWs, a brute-force search or even integer linear programming is impossible to solve within a reasonable time frame. Therefore, this thesis relies on the aforementioned heuristic baselines to evaluate the relative performance gain.

The performance of the proposed algorithm is benchmarked against the following three baseline schemes:

- **Random Selection**: In this baseline, the satellite randomly associates with any visible GW that satisfies the minimum link budget requirements at each decision epoch. This scheme serves as a stochastic lower bound, providing a reference to quantify the performance gain achieved by intelligent decision-making strategies over arbitrary association.

- **Signal-based FLSO**: This scheme adopts the conventional A3 event-based handover rule used in terrestrial cellular networks. A switchover is triggered when the RSRP of a target GW exceeds that of the serving GW by a hysteresis margin. Let $\widetilde{R}_{m,n}(t)$ denote the L3-filtered RSRP from GW m to satellite n at time t. A FLSO from the serving GW m_{serv} to a target GW m_{target} is initiated if:

$$\widetilde{R}_{m_{\text{target}},n}(t) > \widetilde{R}_{m_{\text{serv}},n}(t) + \Delta R_{\text{offset}}, \tag{4.1}$$

  where ΔR_{offset} is the offset margin in dB. THis thesis evaluates two configurations: $\Delta R_{\text{offset}} = 2$ dB (*Set 1*) and $\Delta R_{\text{offset}} = 4$ dB (*Set 2*), representing moderate and conservative switching conditions, respectively. While compatible with 3GPP measurement procedures, this method relies solely on signal strength, ignoring geometric predictability and load balancing.

- **Distance-based FLSO**: This heuristic selects the GW with the minimum Euclidean distance to the satellite, leveraging the deterministic nature of LEO orbits. A FLSO occurs when the distance to a target GW is shorter than that to the serving GW by a specific threshold. The triggering condition is defined as:

$$d_{m_{\text{target}},n}(t) < d_{m_{\text{serv}},n}(t) - \Delta d_{\text{offset}}, \tag{4.2}$$

  where $d_{m,n}(t)$ is the distance between GW m and satellite n, and $\Delta d_{\text{offset}} \geq 0$ is the hysteresis offset in km. this thesis considers

two settings: $\Delta d_{\text{offset}} = 0$ km (*Set 3*) for aggressive closest-GW selection, and $\Delta d_{\text{offset}} = 30$ km (*Set 4*) for conservative switching to suppress ping-pong effects.

- **RL Baselines**: To rigorously evaluate the proposed algorithm and quantify the contribution of each component, this thesis compares it against a suite of value-based RL benchmarks (*Set 5–8*). First, this thesis employs the standard DQN (*Set 5*) and DDQN (*Set 6*) to assess the fundamental performance gains over heuristics and the impact of mitigating Q-value overestimation. Second, to verify the specific benefits of this thesis architectural choices, this thesis examines two ablation variants of the Dueling DDQN: one without PER (*Set 7*) to isolate the effect of sampling efficiency, and another using hard target updates instead of Polyak averaging (*Set 8*) to evaluate training stability.

4.2. Key Performance Indicators

This thesis evaluates seven key performance indicators (KPIs) that together capture FLSO stability, link quality, and GW load balancing. Unless otherwise noted, all metrics are computed per episode and averaged over time and satellites.

FLSO measures how many times, on average, each active connection changes its serving GW during an episode. It reflects the overall switching activity of the FLSO policy. Smaller values indicate a more stable association with fewer disruptions and reduced control overhead.

UFLSO refines this by counting GW switchovers of active connections that are followed by a very short dwell time, shorter than a predefined MTS. Such events correspond to ping-pong or impulsive decisions that do not yield lasting performance gains. This KPI quantifies how many switches are effectively wasteful; a desirable policy minimizes both the total number of switchovers and the fraction that are unnecessary.

ToS measures the average duration that a satellite maintains a continuous connection with a serving GW before a FLSO occurs or the link is terminated. This metric is calculated by averaging the dwell times of all completed and active connections across all satellites and GWs throughout the episode. A larger ToS value signifies a more stable network topology with persistent feeder link associations, which directly translates to reduced signaling overhead for FLSO management and fewer service interruptions, while still permitting necessary switchovers for load balancing or link quality maintenance.

RSRP evaluates the long-term quality of the selected feeder links. At each time step, this thesis observes the L3 filtered RSRP of the GW actually serving each satellite, using the standardized IIR-based parameters in Table 4.1. The feeder link RSRP is then averaged across all satellites and time. Higher values mean that satellites are more consistently attached to GWs providing strong received power, showing that stability is not obtained at the cost of degraded link quality.

Average connections per GW quantifies the mean number of satellites simultaneously served by a single GW at any given time step. This metric provides a direct assessment of the connectivity density per GW.

While maintaining a uniform distribution is generally preferred for load balancing, monitoring this metric is crucial; extremely low values may indicate resource underutilization, whereas excessively high values suggest potential control plane congestion, even if the data plane capacity is not fully saturated.

Total traffic load measures the aggregate data traffic volume processed by a GW, calculated as the sum of the instantaneous traffic demands of all associated satellites. Given that satellite traffic demand is modeled as a stochastic random walk, relying solely on the number of connections is insufficient to capture the actual processing burden. Therefore, evaluating the total traffic load ensures that the FLSO policy effectively distributes the actual data throughput, preventing specific GWs from being overwhelmed by a small number of high-demand satellites.

Jain's fairness index is employed to rigorously evaluate the load balancing performance with respect to the traffic distribution across GWs. Let $L_m(t)$ denote the total traffic load of GW m at time t, and M be the total number of GWs. The index is defined as in 3.9, where the value of $J(t)$ is bounded between $1/M$ and 1. Here, $1/M$ is the worst case, indicating that all traffic is concentrated on a single GW, and 1 is the best case, representing a perfectly uniform load distribution.

4.3. Results and Discussion

Fig. 4.1 presents the convergence of the normalized average reward during the training phase for the proposed dueling DDQN agent and

the baseline schemes Set 5 through Set 8. The results demonstrate the distinct impact of each algorithmic component on learning stability and final performance. The proposed algorithm achieves the highest asymptotic reward and exhibits the most stable convergence trajectory after the initial exploration phase. This confirms that integrating dueling architecture, PER, and Polyak averaging effectively balances the trade-off between multiple objectives maximizing ToS and RSRP while minimizing FLSO frequency and load variance. In contrast, the ablation studies highlight the necessity of each component. Set 7 experiences a catastrophic performance collapse around episode 2,000. This indicates that without prioritized sampling, the agent fails to efficiently learn from sparse but critical negative experiences, leading to temporary policy degradation before slow recovery. Set 8 shows rapid initial learning but saturates at a significantly lower reward level compared to the proposed method. This suggests that the hard update of the target network induces training instability, preventing the agent from fine tuning the policy for optimal satellite selection. Furthermore, while the standard Set 5 and Set 6 show reasonable learning trends, they exhibit higher variance and lower final rewards than the proposed method. Specifically, Set 6 suffers from significant fluctuations, implying that the dueling architecture is crucial for stabilizing value estimation in the complex state space of GWs.

Fig. 4.2 shows the trend of the average FLSO per GW link as the simulation progresses from 100 to 500 steps. Across all algorithms the FLSO count increases approximately linearly with the number of steps, and the relative ordering between aggressive and conservative strategies

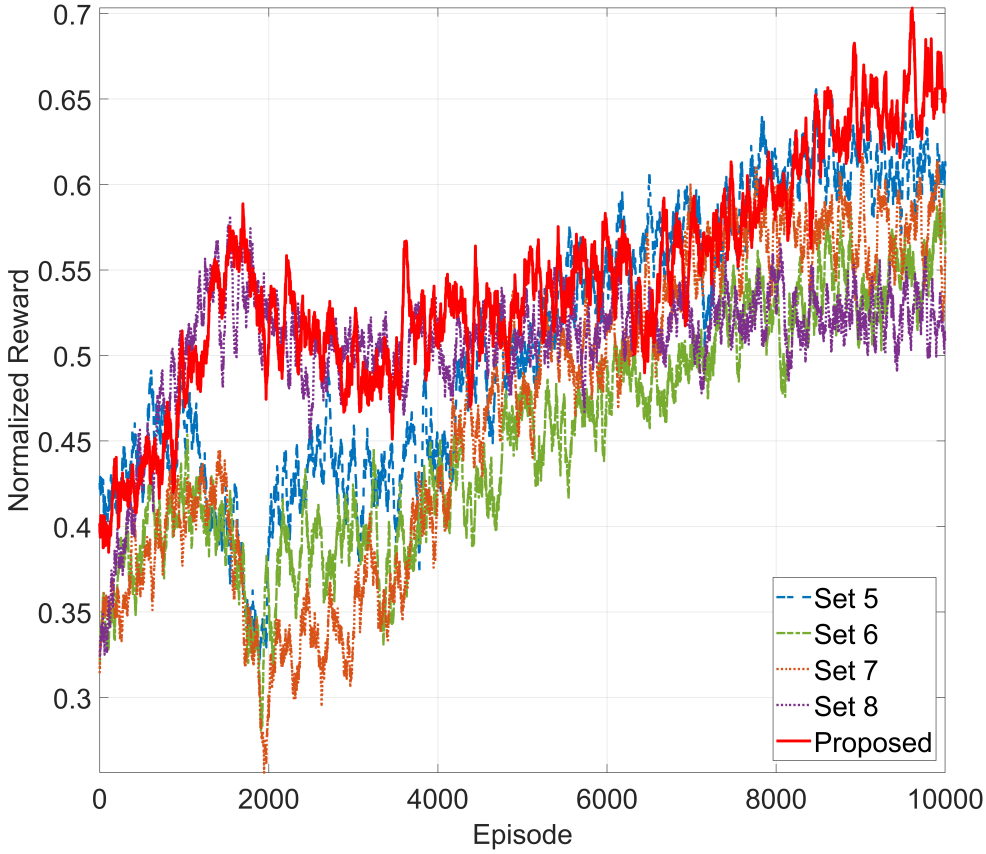


Fig. 4.1. Normalized reward with 50 moving average comparison over episodes for the proposed algorithm and ablation baselines (Set 5–8).

remains consistent at every time scale. Set 1 shows the most aggressive behavior, rising from 4.34 FLSOs at 100 steps to 16.99 at 500 steps, while Set 2 with a larger hysteresis margin grows from 3.17 to 11.55. Set 8 follows a similar trend, rising from 2.92 to 10.52, which confirms that both small hysteresis and naive policy updates lead to unstable connectivity over long time scales. Set 3 and Set 4 show more conservative behavior, evolving from around 2.10 to 8.73 and from 2.10 to 8.40 respectively.

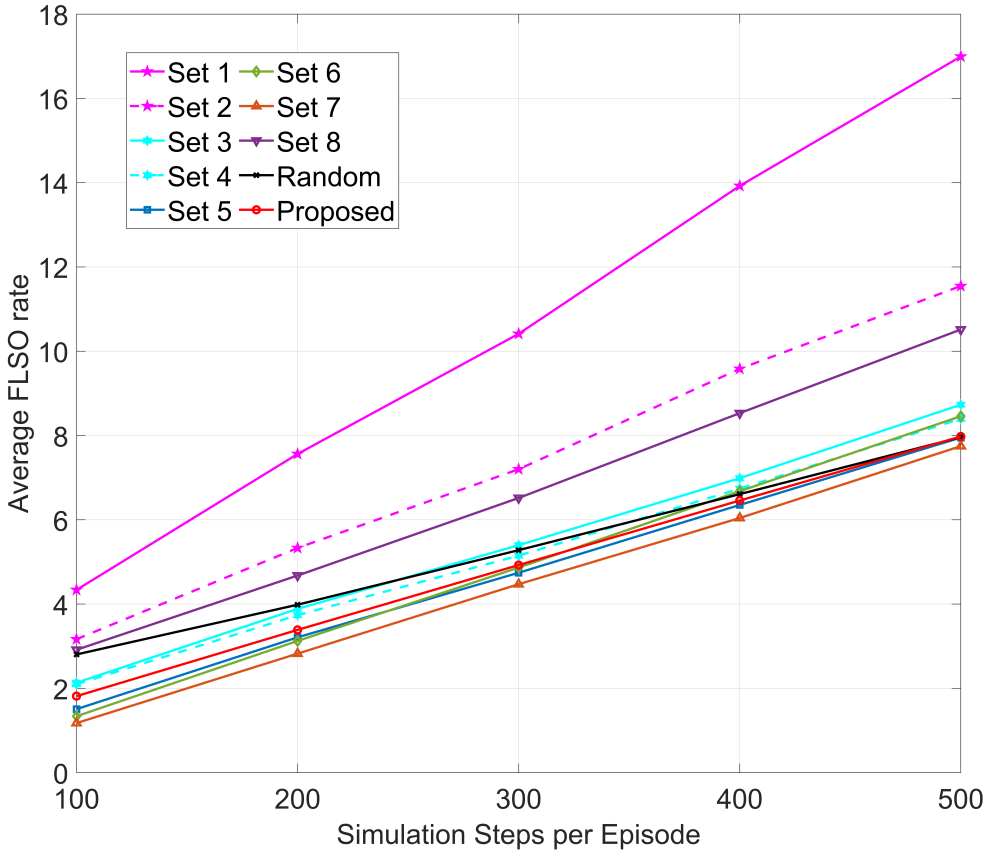


Fig. 4.2. Average FLSO per each GW link for the proposed algorithm and baselines across different simulation steps.

Set 5, Set 6 and Set 7 remain in a comparable range, reaching 7.95, 8.46 and 7.75 FLSOs at 500 steps. The Random policy yields a relatively flat and moderate increase from 2.81 to 7.96 but does so without explicitly optimizing link quality or load balancing. The proposed algorithm consistently maintains a moderate switching rate across all time scales, growing from 1.82 at 100 steps to 7.98 at 500 steps, which is close to or lower than the more conservative baselines.

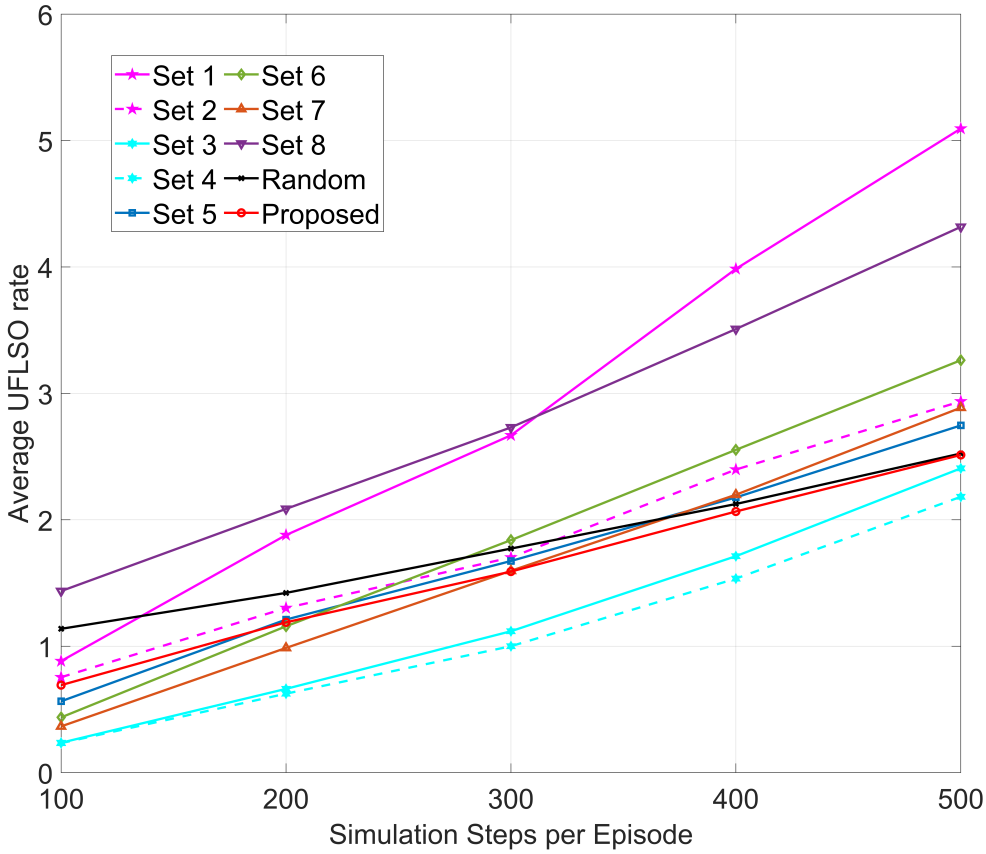


Fig. 4.3. Average UFLSO per each GW link for the proposed algorithm and baselines across different simulation steps.

Fig. 4.3 presents the trend of the average UFLSO count as a function of the simulation length from 100 to 500 steps. For all algorithms, the UFLSO increases almost linearly with the number of steps, and the relative ordering between aggressive and conservative strategies remains stable over time. Set 1 yields the largest UFLSO values, rising from about 0.88 at 100 steps to 5.09 at 500 steps. This shows that the frequent FLSOs generated by a small hysteresis margin translate into a large number of

unstable and ultimately harmful switching operations. Set 2 reduces the UFLSO growth but still reaches 2.94 at 500 steps, which indicates that enlarging the hysteresis margin alone is not sufficient to fully stabilize the FLSO process. Set 3 and Set 4 exhibit clearly more conservative behavior. Their UFLSO values grow from roughly 0.24 at 100 steps to 2.41 and from 0.23 to 2.18 at 500 steps. These curves confirm that limiting switching decisions to geometric criteria suppresses unstable transitions more effectively than purely signal-driven triggering. Set 5, Set 6, and Set 7 occupy an intermediate region. Their final UFLSO values range from 2.75 in Set 5 to 3.26 in Set 6, which is lower than Set 1 and Set 8, but still clearly higher than the distance-based heuristics. Set 8 shows UFLSO growth from about 1.44 to 4.32 and thus remains close to Set 1 in terms of instability. This behavior confirms that naive RL update rules do not provide sufficient regularization of the switching pattern. The Random policy produces UFLSO values between 1.14 and 2.52 over the considered range. The proposed algorithm maintains a consistently low UFLSO rate over the entire simulation horizon. Its UFLSO increases from 0.69 at 100 steps to 2.51 at 500 steps, which places it very close to Set 3 and Set 4.

Fig. 4.4 shows the trend of the total offered service ToS as the number of simulation steps increases from 100 to 500. For all algorithms the ToS grows monotonically with the simulation length, and the relative ordering between the schemes remains consistent over time. Set 1 and Set 2 exhibit the lowest stability region. Their curves start around 30 s to 34.6 s at 100 steps and reach only about 49 s and 63.19 s at 500 steps, which indicates that aggressive reliance on instantaneous signal strength alone does not

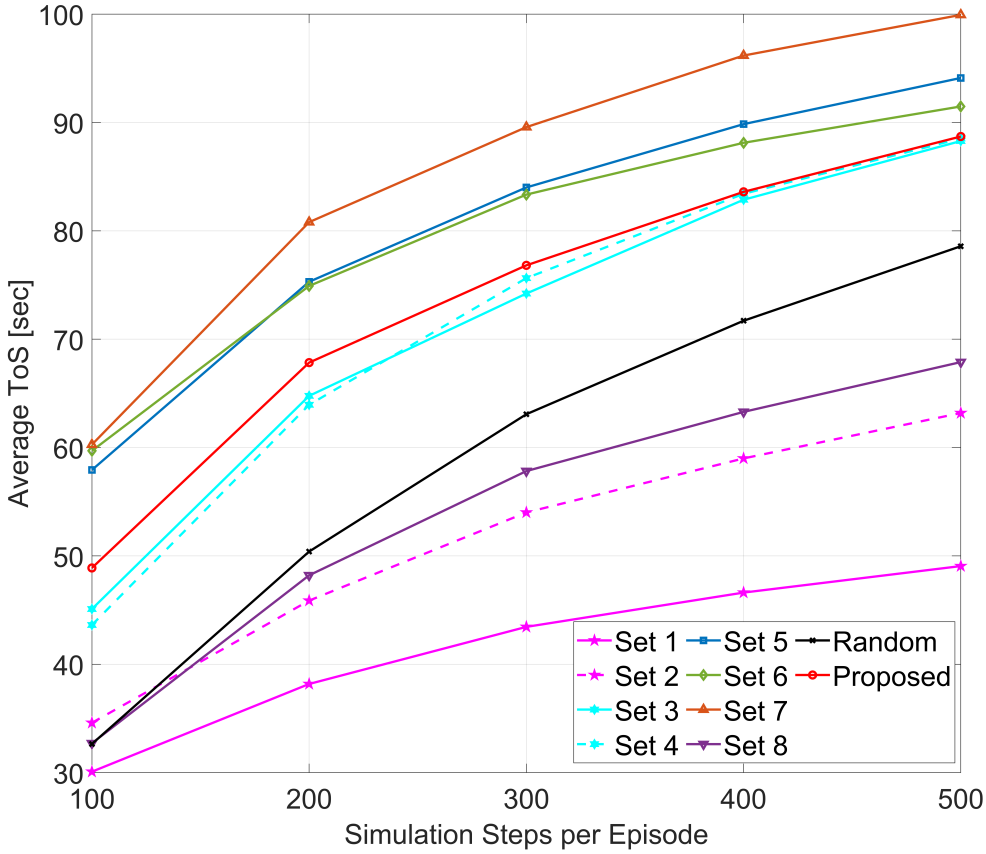


Fig. 4.4. Average ToS per each GW link for the proposed algorithm and baselines across different simulation steps.

fully exploit the available network capacity. Set 3 and Set 4 achieve significantly higher ToS. At 100 steps they already provide more than 44 s of service and continue to increase toward approximately 88.7 s at 500 steps. This demonstrates that even without explicit learning, a carefully designed geometric policy can sustain much higher long-term connection duration than simple signal-driven hysteresis rules. Set 8 lies between these two groups, rising from 32.65 s at 100 steps to about 67.89 s at

500 steps. Set 5, Set 6, and Set 7 form the top stability group. Their ToS values start near 60 s at 100 steps and grow to about 91.48 s, 94.1 s, and 99.9 s at 500 steps. These results confirm that well tuned RL controllers can aggressively exploit the available GW resources to maximize long-term service. The Random policy produces a ToS curve that stays below the RL baselines and distance-based heuristics. It increases from about 62.65 s at 100 steps to roughly 78.58 s at 500 steps. The proposed algorithm closely tracks the best distance-based heuristics. It starts at about 48.89 s at 100 steps and rises steadily to approximately 88.7 s at 500 steps, almost overlapping with Set 3 and Set 4 across the entire simulation horizon.

Fig. 4.5 shows the trend of the average RSRP for all algorithms as the number of simulation steps increases from 100 to 500. Set 1 and Set 2 maintain the highest link quality, with average RSRP values around -55 dB and only very small variations over time. This behavior confirms that aggressive signal-based switching is effective at keeping each satellite attached to the currently strongest GW. Set 3 and Set 4 exhibit clearly lower RSRP levels. Their averages degrade from about -58 dB at 100 steps to around -59.2 dB at 500 steps, which indicates that geometric criteria alone cannot consistently select the strongest links. Set 5, Set 6, and Set 7 and the proposed algorithm form an intermediate group with average RSRP values between roughly -58.4 dB and -58.9 dB at the end of the simulation. Within this group the proposed method closely follows Set 5 and Set 6 and maintains a very stable RSRP trajectory across all time scales. The Random policy starts from about -56.8 dB at 100 steps

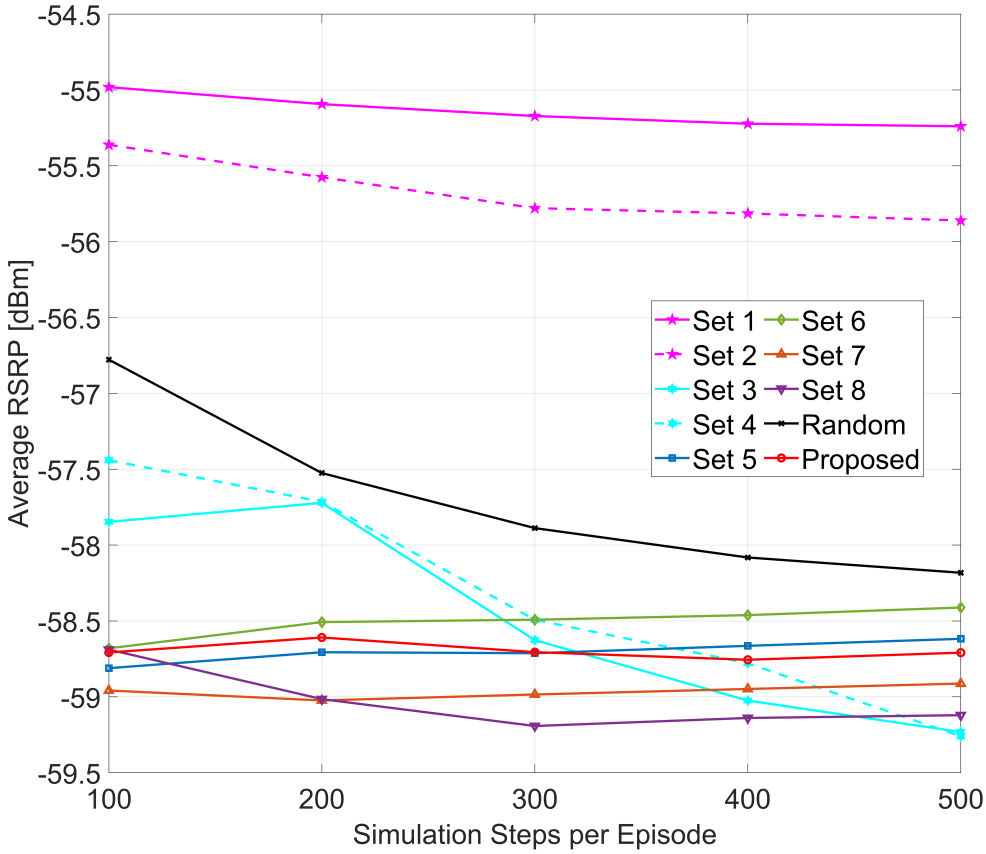


Fig. 4.5. Average RSRP per GW for the proposed algorithm and baselines across different simulation steps.

and gradually converges toward −58.2 dB at 500 steps, which is slightly better than the proposed method in terms of pure signal strength but is achieved without any explicit optimization of load balancing or switching stability. Overall, the proposed controller accepts a moderate loss of approximately 3.5 dB compared with the strongest signal-based schemes in order to obtain the substantial reductions in FLSO and UFLSO reported earlier, while preserving link quality comparable to other RL-based ap-

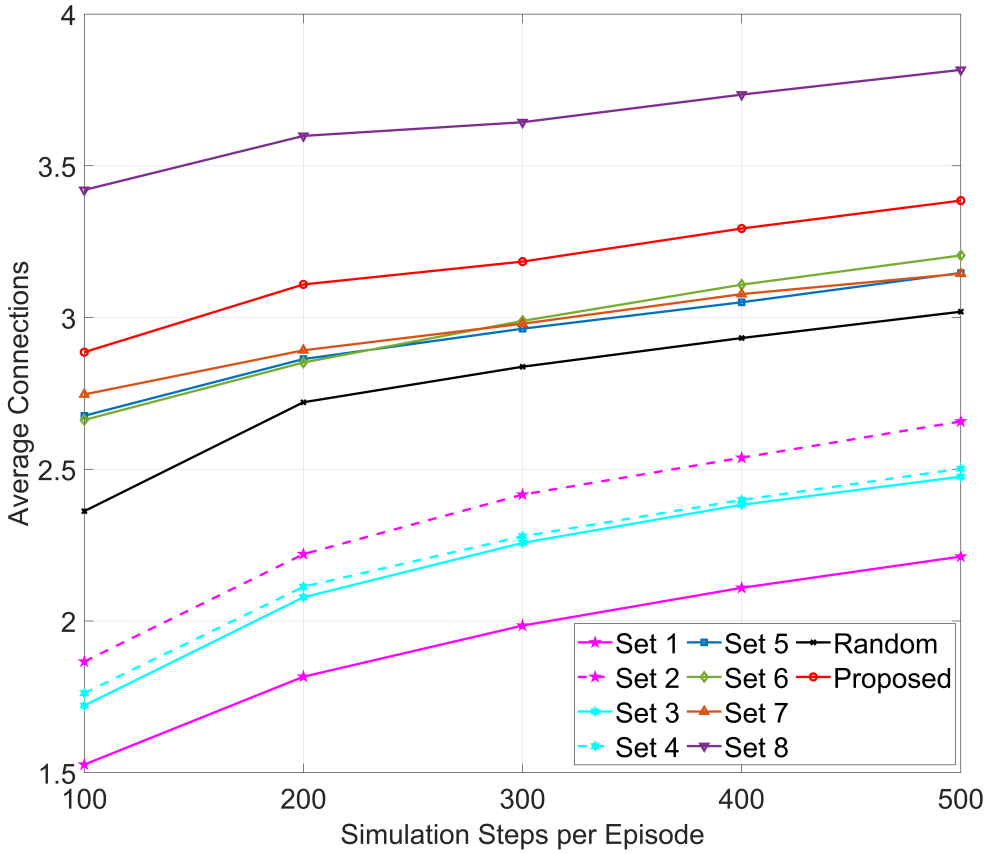


Fig. 4.6. Average number of connections per GW for the proposed algorithm and baselines across different simulation steps.

proaches.

Fig. 4.6 illustrates the trend of the average number of concurrently active satellite connections per GW as the simulation length increases from 100 to 500 steps. For all algorithms this value grows steadily with time since additional FLSO opportunities arise and more links can be activated. Set 1 and Set 2 remain in the lower range of the plot. Their averages increase from about 1.5 to roughly 2.2 in Set 1 and from about

1.9 to roughly 2.7 in Set 2 at 500 steps. Set 3 and Set 4 show a similar behavior, converging to around 2.5 active connections per satellite. These schemes therefore make only limited use of multi-satellite connectivity and primarily rely on one or two strong links at any time. Set 5, Set 6 and Set 7 form a higher connectivity group. Their final averages lie slightly above 3 concurrent GWs, which indicates that the learned policies actively keep multiple links open in order to support load balancing and robustness. Set 8 exhibits the most aggressive behavior, reaching close to 3.8 active connections at 500 steps and thus exploiting almost every available GW around each satellite. The Random policy also maintains a relatively large number of connections, with its average rising from around 2.4 to slightly above 3 across the simulation horizon. The proposed algorithm starts from about 2.9 active connections at 100 steps and grows to approximately 3.4 at 500 steps. This places the proposed controller just below Set 8 and above all other baselines in terms of average connectivity.

Fig. 4.7 shows the average traffic per GW link as the simulation length increases from 100 to 500 steps. For all algorithms the carried traffic grows steadily with time, and the relative order between the schemes is preserved across the entire range. Set 1 and Set 2 occupy the lowest region of the plot. Their average traffic rises from roughly 0.76 to 1.10 Gbps and from 0.94 to 1.33 Gbps at 500 steps, which indicates that relying only on instantaneous signal strength does not fully utilize GW capacity. Set 3 and Set 4 deliver slightly higher traffic than Set 1 and Set 2. Their curves start near 0.81 Gbps and 0.85 Gbps at 100 steps and converge

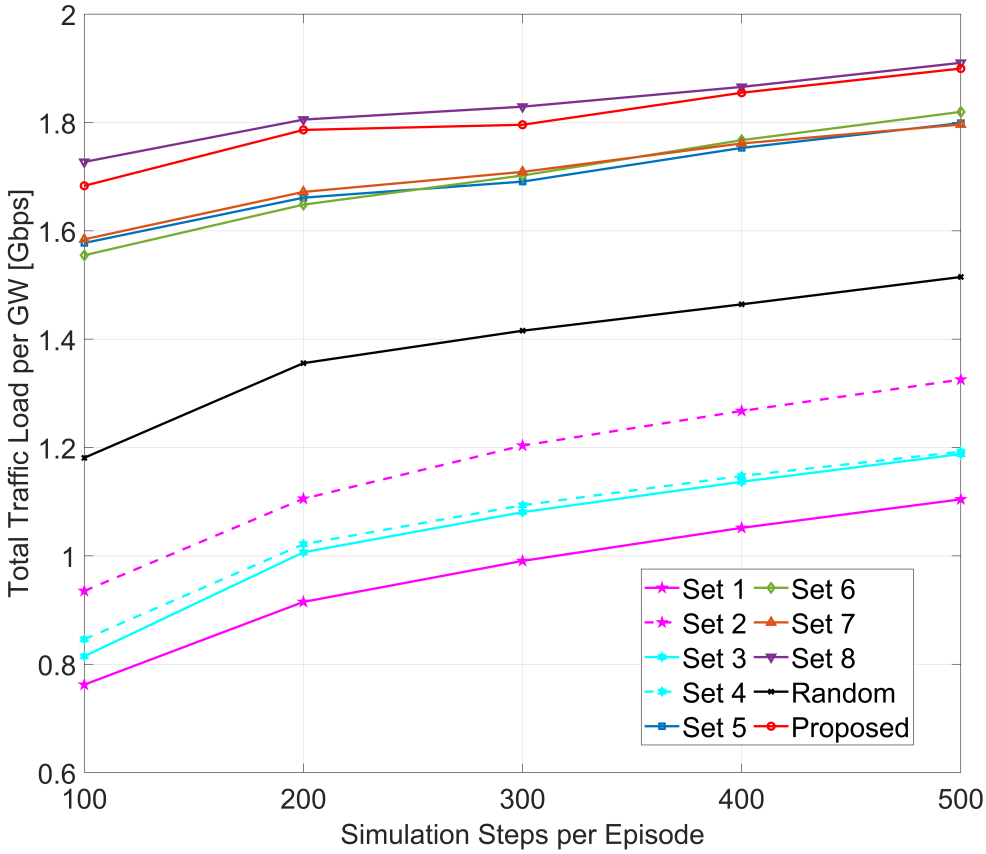


Fig. 4.7. Average total traffic load processed per GW for the proposed algorithm and baselines across different simulation steps.

to about 1.19 Gbps at 500 steps. This suggests that geometric selection of GWs is somewhat more effective in filling the links but still leaves a noticeable margin compared with RL-based controllers. Set 5, Set 6, and Set 7 constitute a higher traffic group. At 500 steps they maintain values between 1.80 Gbps and 1.82 Gbps, which shows that these policies push a large amount of traffic toward the network edge. Set 8 achieves the highest traffic among the baselines. Its average load increases from

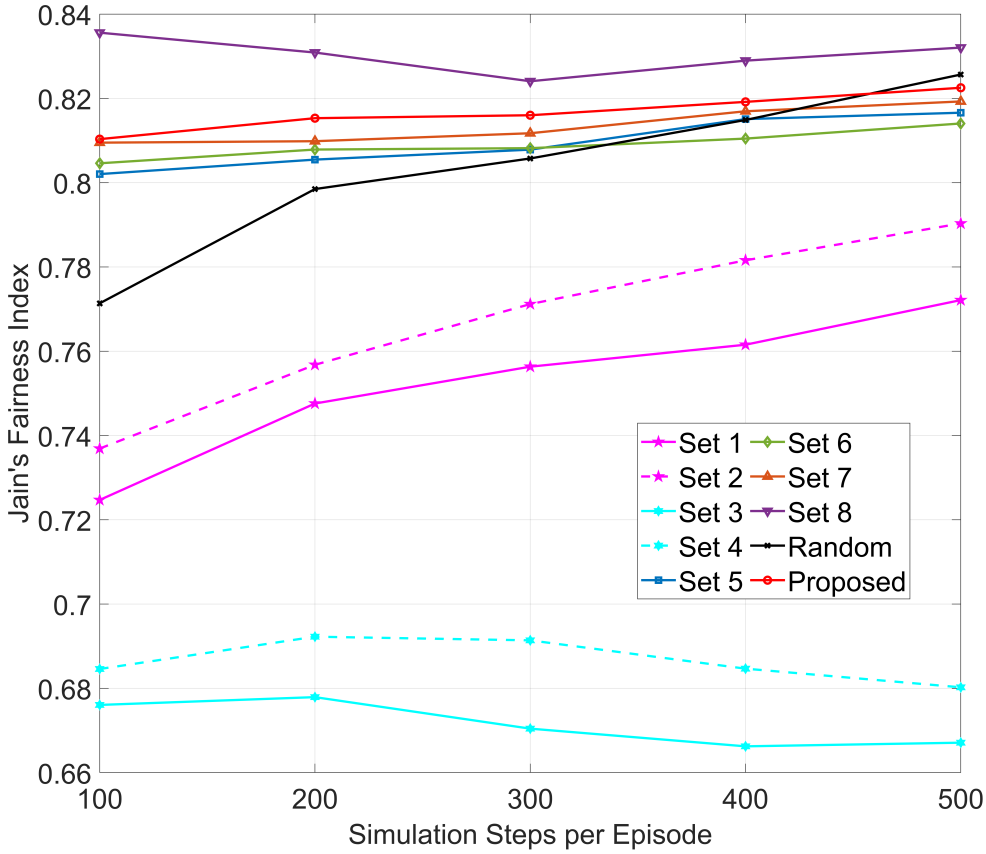


Fig. 4.8. Average Jain's Fairness Index for the proposed algorithm and baselines across different simulation steps.

1.73 Gbps at 100 steps to 1.91 Gbps at 500 steps. The Random policy yields intermediate traffic. It grows from about 1.18 Gbps at 100 steps to 1.51 Gbps at 500 steps, higher than the purely heuristic baselines yet clearly lower than the RL baselines. The proposed algorithm consistently tracks the upper envelope of the RL-based methods. It starts at 1.68 Gbps at 100 steps and reaches 1.90 Gbps at 500 steps, only slightly below Set 8 and above Set 5, Set 6, and Set 7.

Fig. 4.8 illustrates the trend of the fairness index across GW links as the simulation progresses from 100 to 500 steps. For all algorithms the fairness remains relatively stable over time with gradual trends rather than abrupt changes. Set 3 and Set 4 show the lowest fairness among all schemes. Their values stay around 0.67 across the simulation with a slight decrease toward the end. This indicates that purely geometric association tends to concentrate traffic on a subset of GWs and leaves others underutilized. Set 1 and Set 2 achieve a moderate level of fairness. Their indices increase from roughly 0.72 to about 0.79 as the simulation length grows, which confirms that frequent signal-driven switching helps redistribute traffic although the resulting balance is still limited compared with learning-based strategies. Set 5, Set 6 and Set 7 form a high fairness group. Their values stay close to 0.80 throughout the simulation and converge to about 0.82 at 500 steps. Set 8 reaches the highest fairness among all baselines, remaining around 0.83 over the whole time horizon. The Random policy also attains high fairness in a similar range, since randomized association tends to average out load over many GWs. The proposed algorithm starts from a fairness index of about 0.81 and gradually increases to approximately 0.82 at the final step. This places the proposed scheme in the same high fairness cluster as the strongest RL baselines and the best performing baseline Set 8, while clearly outperforming the signal-based and distance-based heuristics.

At 500 simulation steps, the proposed MARL-based FLSO controller achieves the highest overall score among all schemes. Compared with the aggressive and conservative signal-based baselines Set 1 and Set 2, it

reduces FLSO by 53.1% and 30.9%, respectively, and lowers UFLSO by up to 51% relative to Set 1 and 42% relative to the RL baseline Set 8. At the same time, it increases ToS from 63.2 s of Set 2 to 88.7 s, carried traffic from 1.33 Gbps of Set 2 and 1.19 Gbps of Set 4 to 1.90 Gbps, and Jain's fairness index from 0.790 of Set 2 and 0.67 for the distance-based schemes Set 3–4 to 0.823. These gains yield a strictly superior stability, throughput and fairness trade-off to all signal-based, distance-based, and RL baselines, at the cost of only 3.5 dB degradation in average RSRP compared with the strongest signal-based scheme Set 1.

Chapter 5. Conclusion

This thesis investigates a decentralized MARL framework to address GW load balancing in LEO NTN. By modeling the FLSO problem as an MMDP, this thesis develops a multi-agent dueling DDQN algorithm designes to optimize network-wide utility under rapidly time-varying LEO geometries. Extensive simulations demonstrate that the proposed policy achieves the most robust performance trade-off compared to heuristics and standard RL baselines. While signal-based schemes provided high link quality at the cost of excessive switching instability, and distance-based heuristics stabilized connections but led to severe load imbalance, the proposed FLSO successfully balanced these conflicting objectives. It suppressed UFLSOs to levels comparable to conservative geometric strategies while preserving link quality similar to signal-based methods. Crucially, the proposed algorithm significantly improved Jain's fairness index and total throughput, thereby unlocking latent capacity in the ground segment through the intelligent use of multi-GW connectivity. Consequently, this thesis confirms that a lightweight decentralized MARL-based controller can deliver a favorable balance among switching stability, backhaul link quality, and network efficiency.

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